INCH-POUND

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6 April 1992
SUPERSEDING
MIL-C-52464B (ME)
26 July 1982

MILITARY SPECIFICATION

CYLINDER ASSEMBLIES, ACTUATING, LINEAR: HYDRAULIC, FOR BRIDGE AND BRIDGE LAUNCHER

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hydraulic cylinder assemblies for the M48A2, M48A5, and M60A1 bridge launcher-transporters and the hydraulic cylinder assembly for the armored vehicle launched bridge.

1.2 Classification. The cylinder assemblies shall be of the following types and styles, as specified (see 6.2):

Type I	- Hold down, M48A2/M48A5/M60A1
Type II	- Swing, M48A2
Type III	- Scissoring, Bridge Only.
Type IV	- Tongue
Style A	- M48A2
Style B	- M48A5/M60A1
Type V	- Overhead
Style A	- M48A2
Style B	- M48A5/M60A1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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Type VI	- Ejection, M48A5/M60A1
Type VII	- Locking, M48A2/M48A5/M60A1
Type VIII	- Release M48A2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction 2 Way and 4 Way Partial.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment of Painting of Material.
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited Fire-Resistant, Synthetic Hydrocarbon Base.
MIL-C-52950	- Crates, Wood, Open and covered.

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, Waterproofing, with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

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DRAWINGS

MR

TA13211E3020	- Cylinder, Hold-Down.
TA13211E3130	- Cylinder, Locking.
TA13211E3140	- Cylinder, Ejection.
TA13211E7880	- Cylinder, Scissoring, Bridge.
TA13214E0000	- Cylinder, Swing, M48A2.
TA13214E0010	- Cylinder, Overhead, M48A2.
TA13214E0030	- Cylinder, Release, M48A2.
TA13214E0070	- Cylinder, Overhead.
TA13214E0080	- Cylinder, Tongue.
TA13214E0090	- Cylinder, Tongue, M48A2.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents, which are DOD adopted, are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

D 3951	-	Practice for Commercial Packaging.
D 3953	-	Strapping, Flat Steel and Seals.
D 4675	-	Selection and Use of Flat Strapping Material.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, pa 19103).

AMERICAN WELDING SOCIETY, INC. (AWS)

D 1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 550 NW LeJeune Road, and P.O. Box 351040, Miami, FL 33135).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314).

UNIFORM CLASSIFICATION COMMITTEE (UCC).

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The cylinder assemblies or separately procured cylinder components shall be in accordance with the applicable top assembly or drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the cylinders or separately procured cylinder components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Chemical composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests.

3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials, which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished form virgin

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raw materials. The components, pieces and parts incorporated in the cylinder assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the cylinder assembly produced meets all other requirements of this specifications. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the cylinder assembly.

3.4 Cylinders. Each cylinder assembly shall operate at a 3,800 psi sustained working pressure without malfunctioning or leakage. Each cylinder assembly shall withstand the applicable proof pressure specified in table I, without external leakage or visible damage. The rate of internal leakage past the piston of each cylinder assembly at 3,800 psi pressure shall be not more than that specified in table I. All cylinder assemblies shall be filled with hydraulic oil conforming to MIL-H-46170, type 1. All hydraulic cylinder assemblies having the same classification (see 1.2) shall be identical.

TABLE I. Hydraulic Cylinder Requirements

Cylinder	Top Assembly	Application				Rate of Internal Leakage fl-oz/min	Proof Pressure Psi
		Bridge					
		M60A1					
		M48A5					
		M46A2					
Type I	TA13211E3020	X	X	X		4	7,000
Type II	TA13214E0000	X				10	7,000
Type III	TA13211E7880				X	10	7,000
Type IV, Style A	TA13214E0090	X				12	7'000
Type IV Style B	TA13214E0080		X	X		13	7,000
Type V, Style A	TA13214E0010	X				10	7,000
Type V, Style B	TA13214E0070		X	X		13	7,000
Type VI,	TA13214Z3140		X	X		4	7,000
Type VII	TA13211E3130	X	X	X		4	7,000
Type VIII	TA13214E0030	X				N/A	5,000

3.5 Identification marking. The cylinders shall be identified in accordance with MIL-STD-130 and the markings indicated on the drawings.

3.6 Stenciling. The nomenclature, National Stock Number, weight, and cubage shall be stenciled on each cylinder assembly. The letters "NET WT"

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shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4inch high. Stenciling shall conform to MIL-T-704.

3.7 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5):

Item No.	Quantity	Description	Identification
1	1 set	Inspection Equipment for cylinders.	Quality Assurance Pamphlet, P715-124. This pamphlet is included in the Top Assembly.

3.8 Interchangeability. All parts having the same part number shall be functionally and. dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Type I holddown cylinder for M48A2/M48A5/M60A1 launchers. The type I cylinder assembly shall be in accordance with top assembly, TA13211E3020, and as specified herein.

3.10 Type II, swing cylinder for M48A2 launcher. The type II cylinder assembly shall be in accordance with top assembly, TA13214E0000, and as specified herein.

3.11 Type III, scissoring cylinder for the bridge. The type III cylinder assembly shall be in accordance with top assembly, TA13211E7880, and as specified herein.

3.12 Type IV, tongue cylinder.

3.12.1 Style A, for M48A2 launcher. The style A cylinder assembly shall be in accordance with top assembly, TA13214E0090, and as specified herein.

3.12.2 Style B, for M48A5/M60A1 launcher. The style B cylinder assembly shall be in accordance with top assembly, TA13214E0080, and as specified herein.

3.13 Type V, overhead cylinder.

3.13.1 Style A, for M48A2 launcher. The style A cylinder assembly shall be in accordance with top assembly, TA13214E0010, and as specified herein.

3.13.2 Style B for M48A5/M60A1 launchers. The style B cylinder assembly shall be in accordance with top assembly, TA13214E0070, and as specified herein.

3.14 Type VI, ejection cylinder for M48A5/M60A1. Type VI cylinder assembly shall be in accordance with top assembly, TA13211E3140, and as specified herein.

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3.15 Type VII, locking cylinder for M48A2/M48A5/M60A1 launchers. The type VII cylinder assembly shall be in accordance with top assembly, TA1321IE3130, and as specified herein.

3.16 Type VIII, release cylinder for M48A2 launcher. The type VIII cylinder assembly shall be in accordance with top assembly, TA13214E0030, and as specified herein.

3.17 Workmanship.

3.17.1 Welding and welders.

3.17.1.1 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be done by the shielded electric-arc method. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever practicable.

3.17.1.2 Welders or welding operator. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known

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defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. Evidence that any part or component does not comply with the applicable drawings shall be cause for rejection of that part or component. The drawings specify the characteristics requiring (QAP) inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6 and 6.7).

4.1.4 Inspection equipment. Special final inspection equipment to be loaned to the contractor by the Government (see 6.5), is identified in the Quality Assurance Pamphlet listed on the Top Assemblies and 3.7.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article cylinder assembly shall be examined as specified in 4.5.1, as applicable. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cylinder assembly shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each cylinder assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Each cylinder assembly shall be examined as specified herein for the following defects:

101. Any part or component not in accordance with the QAP requirements as shown on the drawings (see 3.1).
102. Any dimension other than those identified by the QAP requirements shown on the drawings not as specified (see 3.1.1).
103. Materials not as specified and analysis and tests not available or missing (see 3.3).
104. Identification marking not as specified (see 3.5).
105. Treating and painting not as specified (see 3.6).

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106. Stenciling not as specified (see 3.6).
107. Components missing or not as specified (see 3.8).
108. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the cylinder assembly (See 3.3.1).
109. Workmanship not as specified (see 3.17).
110. welding and welders not as specified (see 3.17.1.1 and 3.17.1.2').

4.5.2 Tests.

4.5.2.1 Test conditions. Unless, otherwise specified herein, conduct all tests at an ambient temperature of from +40 °F to +90 °F, and oil temperature shall be between +40 °F to +150 °F at pump discharge. Use oil conforming to MIL-H-46170, type 1. Tests shall be conducted in the order listed herein.

4.5.2.2 Proof pressure. Cycle the cylinder assembly with the piston unrestrained through not less than, five complete cycles to demonstrate operation, stroke, and adjustment. Plug the cap-end port and apply a hydraulic pressure of 7,000 psi for a period of 5 minutes at the rod end of each cylinder assembly except for the type VIII cylinder assembly. If the rod moves out of the cylinder assembly during this test, it will be necessary to restrain the rod. Subject the type VIII cylinder assembly to the same test except that the hydraulic pressure shall be reduced to 5,000 psi. The pressure line should be connected to the side port and the cap port left open for one test. For a second test connect the pressure line to the cap end and the side port shall be left open. Evidence of leakage or damage shall constitute failure of this test.

4.5.2.3 Internal leakage.

4.5.2.3.1 Types I, II, III, and IV cylinder assemblies. Connect the type I, II, III, or IV cylinder assemblies as applicable, to a hydraulic system and place the cylinder assembly in a horizontal position with the cap-end port face up. The piston shall be restrained from bearing against the cap end. Insert a graduated, transparent tube of either plastic or glass, and fasten it vertically to the cap-end cylinder port in such a manner as to prevent leakage. One end of the tube shall be open to atmospheric pressure. Apply a 3,800 psi fluid pressure and maintain this pressure in each cylinder assembly, at the rod end. Allow the fluid bypassing the piston to enter and collect in the vertical tube. Measure the rise of fluid level during a period of not less than 3 minutes. Leakage in excess of the rate specified in table I shall constitute failure of this test.

4.5.2.3.2 Type V and VI cylinder assemblies. Test the type V or VI cylinder assemblies as applicable, as specified in 4.5.2.3.1, except that the cap end instead of the rod end of the cylinder assembly shall be pressurized to 3,800 psi. The piston shall be restrained from bearing against the cylinder head. Insert the graduated tube at the rod end of the cylinder assembly. Measure the rise of fluid level during a period of not less than 3 minutes. Leakage in excess of the rate specified in table I shall constitute failure of this test.

4.5.2.3.3 Type VII cylinder assembly. Test the type VII cylinder assembly as specified in 4.5.2.3.1, except that both rod ends of the cylinder assembly shall be tested individually for internal leakage past the restrained piston. Apply a 3,800 psi pressure to one of the rod-end ports, plug the rod-end port, and permit the hydraulic fluid passing the pressurized piston to collect in a tube inserted in the center port. Repeat the test on the other side of the

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cylinder assembly after reversing the pressurized and plugged ports. Leakage in excess of the rate specified in table I shall constitute failure of this test. As an alternate test to the above, the two rod-end ports may be connected to a common line and pressurized to 3,800 psi. If the combined leakage is greater than that specified in table I for one end of the cylinder assembly, both ends of the cylinder assembly shall be tested individually.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. Examine the first article pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The first article pack for level A shall be tested as follows by the applicable test method.

4.6.1.2.1 Boxed cylinder assemblies weighing less than 200 pounds. Boxed cylinder assemblies weighing less than 200 pounds shall be subjected to the free-fall drop test in accordance with FED-STD-101, method 5007.

4.6.1.2.2 Boxed cylinder assemblies weighing more than 200 pounds. Boxed cylinder assemblies weighing more than 200 pounds shall be subjected to the pendulum-impact test in accordance with FED-STD-101, test method 5012.

4.6.1.2.3 Crated Cylinder assemblies weighing more than 1,000 pounds. Crated cylinder assemblies weighing more than 1,000 pounds shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour.

4.6.1.3 Basis for rejection. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the cylinder assemblies shall constitute failure of the first article pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

111.	Cylinder assemblies not fully charged with the specified hydraulic fluid for level A or C (see 5.2.1 and 5.2.2).
112.	Piston rod not fully retracted as specified for level A or C (see 5.2.1 and 5.2.2).

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113. Ports not plugged with steel plugs for level A or C (see 5.2.1 and 5.2.2).
114. Exposed threads of piston rods not protected with steel thread protectors for level A or C (see 5.2.1 and 5.2.2).
115. Preservative not applied as specified for level A or C (see 5.2).
116. Containers not as specified for level A (see 5.3.1).
117. Cylinder assemblies not packed in the container required as specified by weight for level A (see 5.3.1.2).
118. Blocking, bracing, and anchoring not in accordance with the referenced documents for level A (see 5.3.1).
119. Crated cylinder assemblies not protected with a shroud as specified for level A (see 5.3.1.2).
120. Strapping not as specified for level A (see 5.3.1.1).
121. Marking missing, illegible, incorrect, or incomplete for level A or C (see 5.5).
122. Cylinder assemblies not palletized as specified, when required (see 5.4).

5. PACKAGING.

5.1 First article pack. The contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approved by the Government (see 6.10). The first article pack may be prepared utilizing either the first article cylinder assembly or the production cylinder assembly. When the first article model cylinder assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article model cylinder assembly and the production cylinder assembly.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each cylinder assembly shall be filled with the hydraulic fluid specified in 3.4 and the piston rod fully retracted into the cylinder. All ports shall be plugged with steel plugs. Exposed threads of piston rods shall be protected with steel thread protectors. Exposed portions of the piston rod and any other unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservatives. The preservative shall conform to the applicable specification list in and shall be applied in accordance with MIL-P-116.

5.2.2 Level C. Each cylinder assembly shall be preserved as specified for level A except the contractor may use his level C type preservative compound in place of the type P-1 preservative, providing the preservative will afford protection against corrosion and deterioration from the contractor to the initial description.

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5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Cylinder assemblies weighing less than 1,000 pounds. Each cylinder assembly weighing less than 1,000 pounds shall be packed in a close-fitting box conforming to PPP-B-621, Class 2. Style 2 boxes shall be used for cylinder assemblies weighing up to 500 pounds and style 7 boxes with a hood shall be used for cylinders exceeding 500 pounds. The cylinder assembly shall be blocked, braced, and anchored within the box to prevent movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated, and on boxes provided with skids placed adjacent to the skids.

5.3.1.2 Cylinder assemblies weighing more than 1,000 pounds. Each cylinder assembly shall be packed in a crate conforming to MIL-C-52950, style B, Type V, and protected with a waterproof shroud, in accordance with the appendix to the crate specification. Blacking, bracing, and anchoring shall be in accordance with the applicable crate specification and MIL-STD-1186.

5.3.2 Level C. Each cylinder assembly shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.4 Unitization. Type I, VI, VII and VIII cylinder assemblies packed as specified in 5.3.1.1 and 5.3.2 and designated for shipment to production lines shall be palletized when specified (see 6.2). Pallets shall conform to NN-P-71, type and style optional, size 2, group II. The quantity and method of placement on the pallet shall be as recommended by the contractor as determined by the size and configuration of the boxes. The load shall be secured to the pallet with strapping, conforming to ASTM 3953, Type 1 or 2, zinc coated, and ASTM D 4675 width and thickness as appropriate for weight of the load.

5.5 Marking In addition to any special marking specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The cylinder assemblies are intended to be used as components of production bridges and production M48A5/M60A1 Bridge Launchers and as replacement/spare parts for M48A2/M48A5/M60A1 Bridge Launchers and bridges currently in the army inventory.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type and style required (see 1.2).
d. Date of issue of DODISS applicable to this contract and exceptions thereto (see 2.1.1).

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e. Time frame required for submission of first article model (see 3.2).
f. When Government-loaned property is not required (See 3.7).
g. Time frame required for submission of the first article pack (see 5.1).
h. Level of preservation and packing required (see 5.2 and 5.3).
i. When palletization is required (see 5.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Contractors. The contracting officer shall stipulate in the contract that all hydraulic cylinders furnished under terms of any one contract shall be manufactured by the same contractor.

6.5 Government-loaned property. The contracting officer should arrange to loan the inspection equipment specified in 3.7 when applicable.

6.6 Quality assurance provision (QAP). The contracting officer should require the contractors to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.7 Consideration of data requirements. The copies of analyses and tests reports for all materials should be considered when this specification is applied on a contract.

6.8 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference paragraph	DID Number	DID Title
3.17.12	DI-MISC-80876	Welding Procedure Qualification Test Report
3.17.12	DI-MISC-80875	Welding Procedures

6.9 Definition.

6.9.1 Quality assurance provision (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10 First article. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the cylinders in accordance with this specification.

6.11 Subject terms (key word) listing.

Bridge launchers
Bridge launcher-transporters
Hydraulic cylinder assemblies

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

Custodian:
Army - ME

Review activity:
Army - AT

Preparing activity:
Army - ME

Project No. 5420-A384

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